1. Define the following terms:

a) Specificity b) Sensitivity c) FRR d) FAR

e) ROC curve

Ans: The definitions are given below:

a) Specificity: It is the proportion of true negatives that are correctly predicted by a model. A high specificity means that a model correctly identifies most of the negative samples, while a low specificity means that a model mislabels a lot of negative samples as positive. It can be calculated as,

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Where,

TN = True Negative

FP = False Positive

b) Sensitivity: Sensitivity is a measure of how well a machine learning model can detect positive instances. It is also known as True Positive Rate (TPR) or recall. Sensitivity is used to evaluate model performance because it allows us to see how many positive instances the model was able to correctly identify. The higher the sensitivity, the better a model in identifying positive instances. It can be calculated using the following formula:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

where,

TP = True Positive

FN = False Negative

c) FRR: False Rejection Rate (FRR) is the probability that a system rejects an authorized person incorrectly. It is a measure of performance evaluation of a biometric system. It is calculated by dividing the number of false rejections by the number of genuine samples.

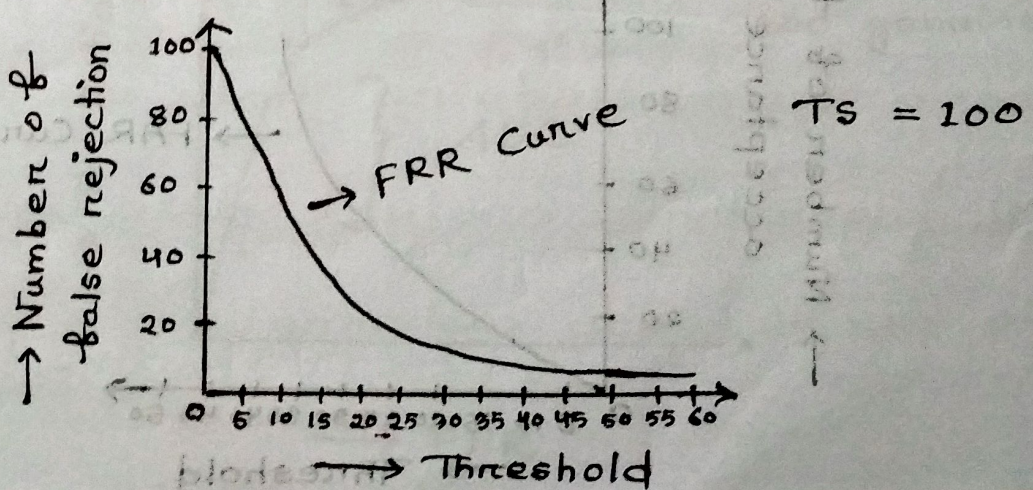
$$FRR = \frac{FR}{TS}$$

Here,

FR = Number of false rejections

TS = Total genuine samples

We can FRR curve by plotting FRR values in graph in different threshold values.



d) FAR: False Acceptance Rate (FAR) is the probability that a system incorrectly accepts an unauthorized person. It is calculated by dividing the number of false acceptances by the number of fake identification samples.

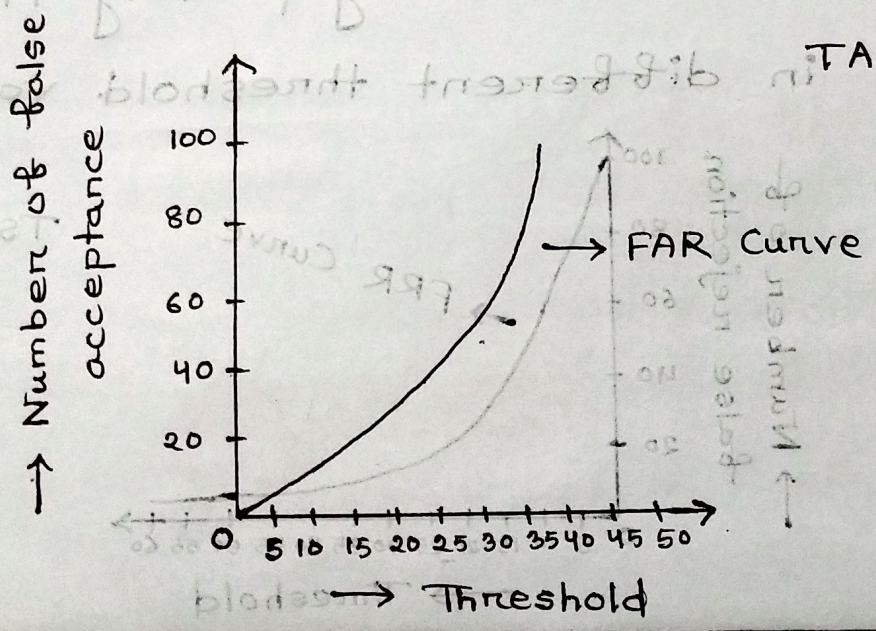
$$FAR = \frac{FA}{TA}$$

Here,

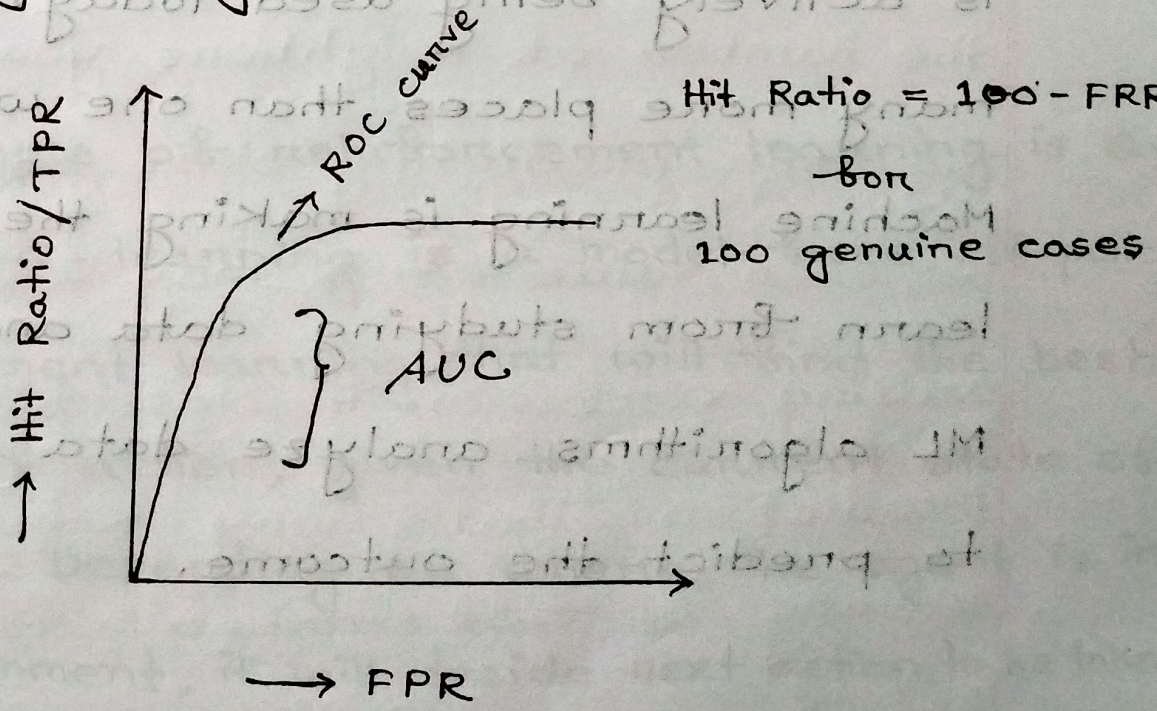
FA = Number of false acceptance

TA = Total number of fake identification attempts

We can draw FAR curve by plotting FAR values in different threshold in a graph:



e) ROC Curve : Receiver Operating Characteristic is a graph showing the performance of a classification model at all classification thresholds. This curve plot two parameters : True Positive Rate, False Positive Rate. The ROC curve shows the trade-off between sensitivity (TPR) and specificity (1-FPR) or Hit ratio. Area under the curve (AUC) is the entire area beneath the ROC curve. AUC represents the degree of separability. It tells how much a model is capable of distinguishing between classes.



2. What is Machine Learning?

Ans: Machine Learning: Machine Learning is the field of study that gives computers the capability to learn without being explicitly programmed. ML is one of the most exciting technologies that one would even come accross. Machine Learning is a step into the direction of Artificial Intelligence (AI). It gives the computers the ability to learn that makes it more similar to humans. Machine learning is actively being used today, perhaps in many more places than one would expect. Machine learning is making the computer learn from studying data and statistics. ML algorithms analyze data and learns to predict the outcome.

2. What is reinforcement learning ? Give an example.

Ans: Reinforcement Learning: Reinforcement learning is a machine learning training method based on rewarding desired behaviors and discouraging undesired ones. In general, a reinforcement learning agent is able to perceive and interpret its environment, take actions and learn through trial and error. There are four types of reinforcement : positive reinforcement, negative reinforcement, extinction, and punishment.

Example:

One example of reinforcement learning is Q-learning. Q-learning is a model-free, off-policy reinforcement learning that will find the best course of action, given the current state of an agent. Depending on where the agent is in the environment, it will decide next action to be taken.

4. Why value of 'K' should be odd in K-nearest neighbors algorithm?

Ans: KNN is a majority voting Algorithm. Each instance in essance votes ~~for~~ their classes and the class with the most votes is taken as the prediction. If you are using K and you have an even number of classes, it is a good idea to choose a K value with an odd number to avoid tie. The K in the KNN algorithm, represent the number of neighbors you are comparing for predicting class of new instances. So, even value of K may lead to a tie arising confusion in classification or prediction. That's why K in KNN is usually an odd number.

5. Differentiate between supervised and unsupervised learning.

Ans: The differences are given below:

Supervised Learning	Unsupervised Learning
1. Algorithms are trained using labeled data.	1. Algorithms are trained using unlabeled data.
2. Supervised learning model predicts the output.	2. Unsupervised learning model finds the hidden pattern in data.
3. Input data is provided to the model along with output.	3. Only input data is provided to the model.
4. Supervised learning needs supervision to train the model.	4. Doesn't need any supervision to train the model.
5. Supervised learning can be categorized in classification and Regression problems.	5. Unsupervised learning can be classified in Clustering and Associations problem.

6. What is overfitting?

Ans: Overfitting: Overfitting is a situation that occurs when a machine learning or deep neural network model performs significantly better for training data than it does for new data. The most common causes of overfitting include the following.

→ The data is used to train the model is dirty and contains large amounts of noise.

→ The model has a high variance with data points that are very spread out from the statistical mean and from each other.

→ The size of the training dataset is too small.

→ The model was created by using a subset of data that doesn't accurately represent the entire data set.

k-fold cross-validation detects overfitting.

7. Write the formula of Weighted Euclidean distance Matrix.

Ans: The standanized Euclidean distance between two I-dimensional vectors can be written as:

$$d_{x,y} = \sqrt{\sum_{i=1}^{I} \left(\frac{x_i}{s_i} - \frac{y_i}{s_i} \right)^2}$$

where,

s_i is the sample standard derivation of the i-th variable.

The formula can be re-written as follows:

$$d_{x,y} = \sqrt{\sum_{i=1}^{I} \frac{1}{s_i^2} (x_i - y_i)^2}$$

$$= \sqrt{\sum_{i=1}^{I} w_i (x_i - y_i)^2}$$

where,

$w_i = \frac{1}{s_i^2}$ is the inverse of ith variance

We think of w_i as a weight attached to the i-th variable.

8. Describe the procedures of a verification system.

Ans: Data flow diagram of a verification system is given below:

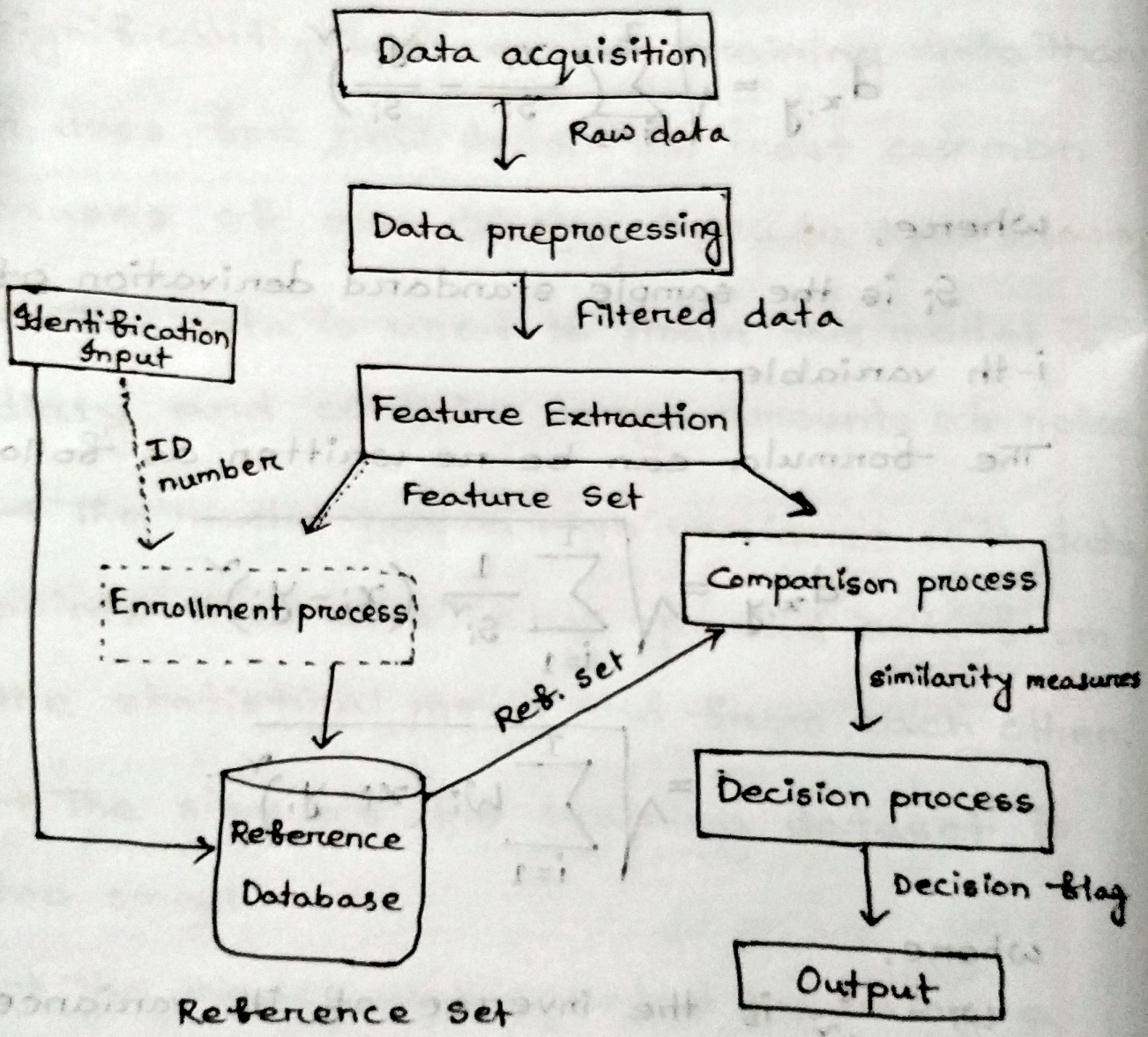


Fig: Data Flow Diagram of a verification system

9. Write the K-means clustering algorithm.

Ans: The K-means algorithm is given below:

begin initialize $n, c, \mu_1, \mu_2, \ldots, \mu_c$

do classify n samples according to nearest μ_i

recompute μ_i

until no change in μ_i

return $\mu_1, \mu_2, \ldots, \mu_c$

end

The computational complexity of the algorithm: $O(ndcT)$

where,

d = number of features

T = number of iterations

c = number of clusters

n = number of samples

10. Write a program to implement Edit distance algorithm. Test the program with 'exercised' and 'executed'.

Ans: A python program to implement edit distance algorithm is given below:

```
def edit_distance(s1, s2):
    m, n = len(s1), len(s2)
    dp = [[0] * (n+1) for in range(m+1)]

    for i in range(m+1):
        dp[i][0] = i

    for j in range(n+1):
        dp[0][j] = j

    for i in range(1, m+1):
        for j in range(1, n+1):
            if s1[i-1] == s2[j-1]:
                dp[i][j] = dp[i-1][j-1]
```

```
        else:
            dp[i][j] = 1 + min(dp[i-1][j], dp[i][j-1],
                               dp[i-1][j-1])

    return dp[m][n]

def main():
    s1 = input("Enter first string:")
    s2 = input("Enter second string:")
    print("The edit distance between {} and {}
          is: {}".format(s1, s2, edit_distance(s1,s2)))

if __name__ == '__main__':
    main()
```

Sample Input and Output:

```
Enter first string: exercised
Enter second string: executed

The edit distance between exercised and executed
is: 3.
```

11. Distinguish between feed forward network and feedback / recurrent network.

Ans:

A feed-forward neural network is a type of artificial neural network where the information flows in only one direction, from the input layer through one or more hidden layers to the output layer. There are no loops in the network, and the information moves forward only. Feed-forward neural networks are used in variety of applications such as pattern recognition, image processing and natural language processing.

On the other hand, a recurrent or feedback network is a type of artificial neural network where the output of a particular layer is fed back into the input of the same or a previous layer.

Recurrent networks have loops in the network, which allows the network to store information over time. This makes them particularly useful for time series prediction and sequential data processing, such as in speech recognition or natural language processing.

One key difference between the two is that feedforward networks can only process a fixed-size input at a time, while recurrent neural networks can handle input of arbitrary size. This is because recurrent networks maintain a state that can be updated at each time step, allowing them to process inputs of variable length.

Another important difference is that feedforward networks don't have memory, while recurrent networks do. The feedback loops

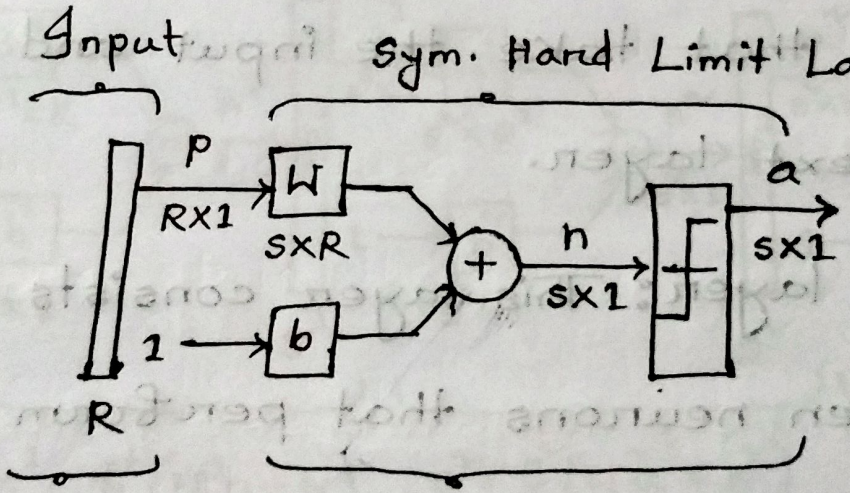
in recurrent networks allow them to maintain and update a memory of the input sequence, which makes them particularly effective for tasks that require context or temporal information.

The main difference between them lies in their ability to handle sequential or temporal data. Feedforward networks are better suited to problems where the input data is fixed in size, while recurrent networks are better suited to problems that require memory and context over time.

Feedforward networks are also known as Multi-Layer Perceptrons (MLP). Recurrent neural networks include popular architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU).

12. Design a single layer perceptron.

Ans: A single layer perceptron consists of input nodes, a layer of output nodes and a set of weights connecting the two layers. Each input node represents a feature or attribute of the input data, and each output node represents a class or category to which the input data belongs. The weights determine the strength of the connections between the input and output nodes, and are initially assigned random values.



$a = hardlims(wp+b)$

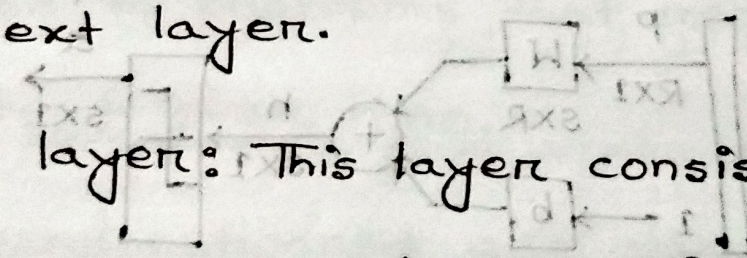
Fig: Single-layer Perceptron

13. Design a multi-layer perceptron.

Ans: An MLP is a type of feed-forward network, which consists of an input layer, one or more hidden layers, and an output layer. Each layer consists of one or more neurons, which apply an activation function to the weighted sum of the inputs.

Here is the basic design for an MLP:

1. Input layer: This layer consists of input neurons that take the input and pass it to the next layer.

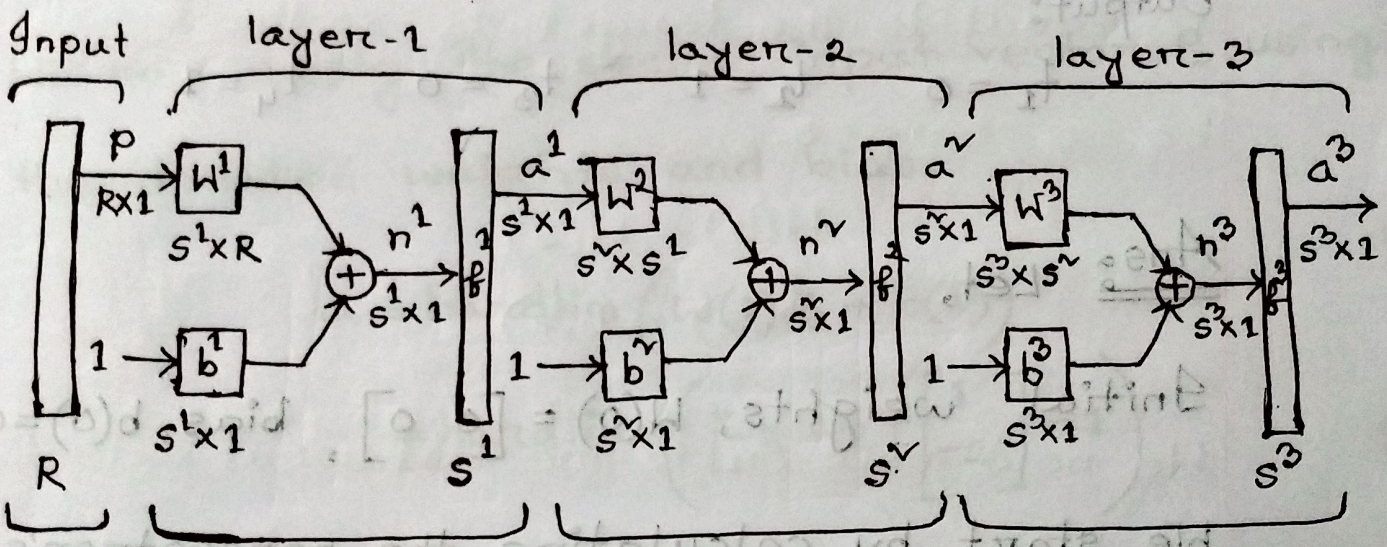


2. Hidden layer: This layer consists of multiple hidden neurons that perform the computations on the input data. The number of hidden layers depends on problem complexity.

3. Output layer: This layer consists of output neurons that produces the final output.

4. Activation Function: Each neuron in the hidden and output layer is associated with an activation function that introduces non-linearity into the model. Common activation function is ReLU, sigmoid and tanh.

5. Loss Function: The loss function measures the difference between the predicted outputs and the true outputs.



$$a^1 = f^1(W^1p + b^1) \quad a^2 = f^2(W^2a^1 + b^2) \quad a^3 = f^3(W^3a^2 + b^3)$$

$$a^3 = f^3(W^3f^2(W^2f^1(W^1p + b^1) + b^2) + b^3)$$

Fig: Three-layer Network (MLP)

14. Solve the following classification problem with the perceptron rule. Apply each input vector in order, for as many repetition as it takes to ensure that the problem is solved. Draw a graph of the problem only after you have found a solution.

Input:

$$P_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad P_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad P_3 = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \quad P_4 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Output:

$$t_1 = 0 \quad t_2 = 1 \quad t_3 = 0 \quad t_4 = 1$$

Ans: Let,

Initial weights, $W(0) = [0 \ 0]$, bias, $b(0) = 0$

We start by calculating the perceptron's output a for the first input vector P_1, using the initial weights and bias.

$$a = \text{hardlim}\left(W(0)P_1 + b(0)\right)$$

$$= \text{hardlim}\left(\begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 0 \right) = \text{hardlim}(0)$$

$$= 1$$

The output a doesn't equal the target value t_1, so we use the perceptron rule to find new weights and biases based on the error.

$$e = t_1 - a = 0 - 1 = -1$$

$$W(1) = W(0) + e p_1^T = \begin{bmatrix} 0 & 0 \end{bmatrix} + (-1) \begin{bmatrix} 2 & 2 \end{bmatrix} = \begin{bmatrix} -2 & -2 \end{bmatrix}$$

$$b(1) = b(0) + e = 0 + (-1) = -1$$

We now apply the second input vector P_2, using the updated weights and bias.

$$a = \text{hardlim}\left(W(1) P_2 + b(1) \right)$$

$$= \text{hardlim}\left(\begin{bmatrix} -2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} - 1 \right)$$

$$= \text{hardlim}(1) = 1$$

This time the output a is equal to the target t_2. Application of the perceptron rule will not result in any changes.

$$W(2) = W(1)$$

$$b(2) = b(1)$$

We now apply the third input vector.

$$a = \text{hardlim}\left(W(2)P_3 + b(2)\right)$$

$$= \text{hardlim}\left(\begin{bmatrix}-2 & -2\end{bmatrix}\begin{bmatrix}-2\\2\end{bmatrix} - 1\right)$$

$$= \text{hardlim}(-1) = 0$$

The output in response to input vector P_3 is equal to the target t_3, so there will be no changes.

$$W(3) = W(2)$$

$$b(3) = b(2)$$

We now move on to the last input vector P_4.

$$a = \text{hardlim}\left(W(3)P_4 + b(3)\right)$$

$$= \text{hardlim}\left(\begin{bmatrix}-2 & -2\end{bmatrix}\begin{bmatrix}-1\\1\end{bmatrix} - 1\right)$$

$$= \text{hardlim}(-1)$$

$$= 0$$

This time the output a doesn't equal to the target t_4. The perceptron rule will result in a new set of values for weights and bias.

$$e = t_4 - a = 1 - 0 = 1$$

$$W(4) = W(3) + eP_4^T = [-2 \ -2] + (1)[-1 \ 1] = [-3 \ -1]$$

$$b(4) = b(3) + e = -1 + 1 = 0$$

We now must check the first vector P_1 again.

This time the output a is equal to the associated target t_1.

$$a = \text{hardlim}(W(4)P_1 + b(4))$$

$$= \text{hardlim}\left([-3 \ -1]\begin{bmatrix}2\\2\end{bmatrix} + 0\right)$$

$$= \text{hardlim}(-8) = 0$$

Therefore there are no changes.

$$W(5) = W(4)$$

$$b(5) = b(4)$$

The second presentation of P_2 results in an error and therefore a new set of weight and bias values.

$$a = \text{hardlim}\left(W(5)P_2 + b(5)\right)$$

$$= \text{hardlim}\left(\begin{bmatrix}-3 & -1\end{bmatrix}\begin{bmatrix}1\\-2\end{bmatrix} + 0\right)$$

$$= \text{hardlim}(-1) = 0$$

Here are the new values:

$$e = t_2 - a = 1 - 0 = 1$$

$$W(6) = W(5) + eP_2^T = \begin{bmatrix}-3 & -1\end{bmatrix} + 1\begin{bmatrix}1 & -2\end{bmatrix} = \begin{bmatrix}-2 & -3\end{bmatrix}$$

$$b(6) = b(5) + e = 0 + 1 = 1$$

Cycling through each input vector once more results in no errors.

$$a = \text{hardlim}\left(W(6)P_3 + b(6)\right) = \text{hardlim}\left(\begin{bmatrix}-2 & -3\end{bmatrix}\begin{bmatrix}-2\\2\end{bmatrix} + 1\right)$$

$$= \text{hardlim}(-1) = 0 = t_3$$

$$a = \text{hardlim}\left(W(6)P_4 + b(6)\right) = \text{hardlim}\left(\begin{bmatrix}-2 & -3\end{bmatrix}\begin{bmatrix}-1\\1\end{bmatrix} + 1\right)$$

$$= \text{hardlim}(0) = 1 = t_4$$

$$a = \text{hardlim}\left(W(6)P_1 + b(6)\right) = \text{hardlim}\left(\begin{bmatrix}-2 & -3\end{bmatrix}\begin{bmatrix}2\\2\end{bmatrix} + 1\right)$$

$$= \text{hardlim}(-9) = 0 = t_1$$

$$a = \text{hardlim}\left(W(6)P_2 + b(6)\right) = \text{hardlim}\left(\begin{bmatrix}-2 & -3\end{bmatrix}\begin{bmatrix}1\\-2\end{bmatrix} + 1\right)$$

$$= \text{hardlim}(5) = 1 = t_2$$

Therefore, the algorithm has converged. The final weights and bias are:

$$W = \begin{bmatrix}-2 & -3\end{bmatrix} \quad \text{and} \quad b = 1$$

Now, we can graph the training data and the decision boundary of the solution. The decision boundary is given by:

$$n = Wp + b = w_{1,1}p_1 + w_{1,2}p_2 + b$$

$$= -2p_1 - 3p_2 + 1 = 0$$

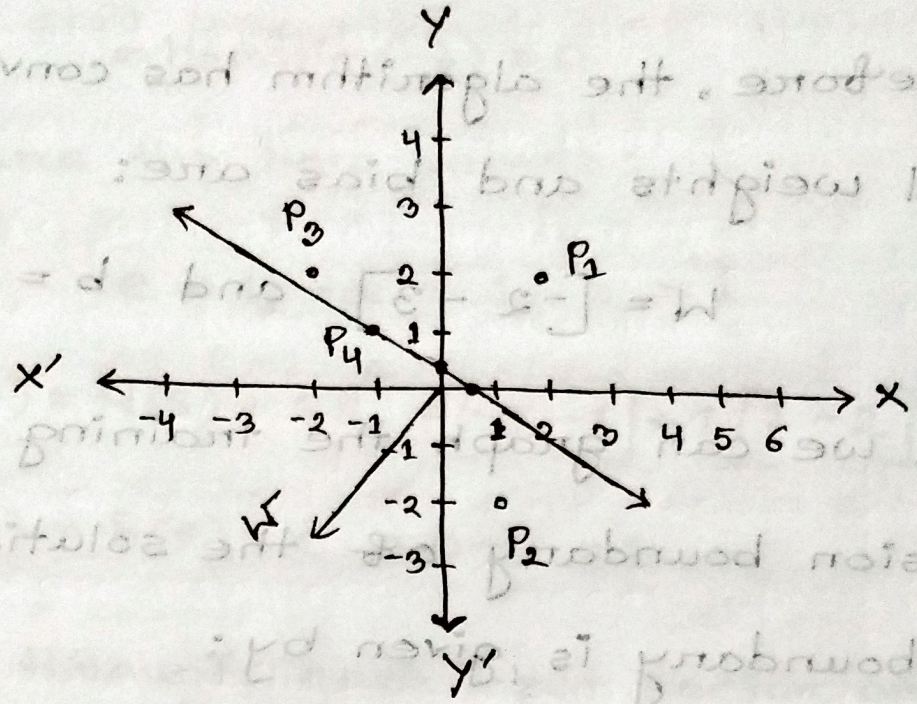
To find the p_2 intercept of the decision boundary, set $p_1 = 0$:

$$p_2 = -\frac{b}{w_{1,2}} = \frac{-1}{-3} = \frac{1}{3} \quad \text{if} \quad p_1 = 0.$$

To find p_1 intercept, set $p_2 = 0$:

$$p_1 = -\frac{b}{w_{1,1}} = -\frac{1}{-2} = \frac{1}{2} \quad \text{if} \quad p_2 = 0.$$

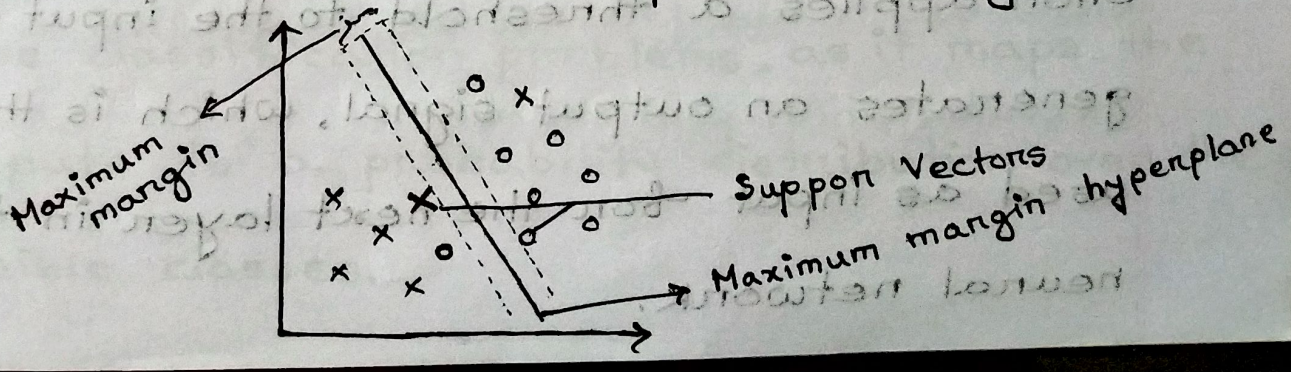
The resulting decision boundary is given below:



The decision boundary fall across one of the training vectors. This is acceptable, since the hard limit function returns 1 when given an input of 0, and the target for the vector in question is indeed 1.

15. What is "support vector" in SVM classification algorithm?

Ans: In Support Vector Machine (SVM), a support vector is a data point that is closest to the decision boundary that separates the different classes. The decision boundary is defined by a hyperplane that maximally separate the classes. The support vectors are the data points that lie on the margin of this hyperplane, meaning that they have the smallest distance to the hyperplane. SVM algorithm identifies the support vectors by solving the optimization problem that involves maximizing the margin and finding the best hyperplane that can separate the classes with smallest possible margin violation.



16. What is activation function?

Ans: An activation function is a mathematical function that is applied to the output of a neural network layer or node within that layer. The activation function is used to introduce non-linearity into the output of a neural network. Without an activation function, a neural network would simply be a linear regression model, which is limited in its ability to learn complex relationships between inputs and outputs.

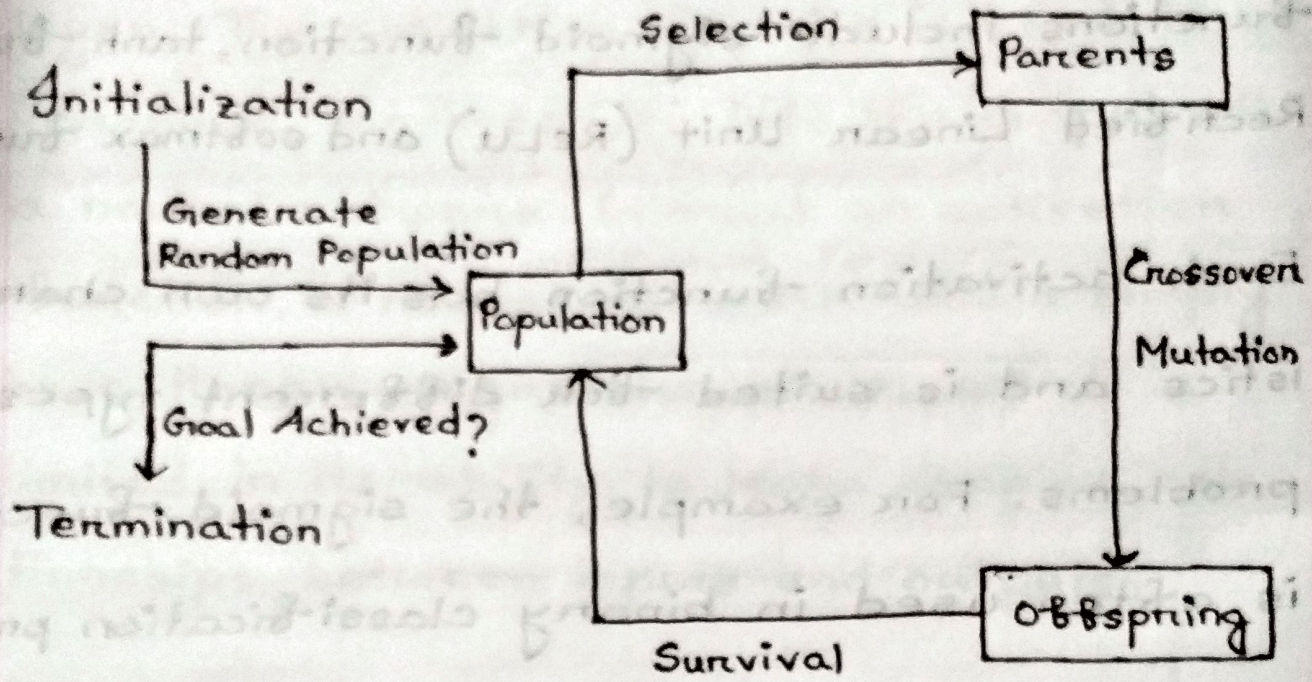
The activation function takes in the weighted sum of the inputs and biases, calculates a value, and passes it onto the next layer of the network. The activation function applies a threshold to the input and generates an output signal, which is then used as input for the next layer in the neural network.

There are many different activation functions available, and the choice of activation function can have a significant impact on the performance of the neural network. Some popular activation functions include sigmoid function, tanh function, Rectified Linear Unit (ReLU) and softmax function.

Each activation function has its own characteristics and is suited for different types of problems. For example, the sigmoid function is often used in binary classification problems while the ReLU function is used in deep neural networks for image and speech recognition tasks. The softmax activation function is commonly used in the output layer in multi-class classification problems, as it maps the outputs to a probability distribution over the possible classes.

17. Draw the flowchart of genetic algorithm.

Ans: The flowchart of evolutionary process is given below:



where,

population : a set of solutions (or individuals/ chromosomes)

Generation: the process of evaluation, selection, recombination and mutation.

18. Suppose that the machine learning task is to cluster the following students' CGPA into three clusters i.e. K = 3,

A	B	C	D	E	F
3.45	3.78	2.98	3.24	4.0	3.9

Use K-means algorithm for the tasks and let the initial centers be A, C and E. Find the final three clusters.

Ans:

step 1: Initialize the cluster centers

As given in the problem, we start by initializing the cluster centers to A, C, and E. So our initial cluster centers are:

Cluster-1: center, A = 3.45

Cluster-2: center, C = 2.98

Cluster-3: center, E = 4.0

step 2: Assign each point to its closest cluster center.

Next, we assign each student's CGPA to the closest cluster center based on the Euclidean distance between the data point and the cluster center. The following equation is used to calculate Euclidean distance between point (x_1, y_1) and (x_2, y_2).

$$\text{Distance, } d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Student	CGPA	Distance from Cluster-1	Distance from Cluster-2	Distance from Cluster-3	Assigned Cluster
A	3.45	0.00	0.47	0.55	1
B	3.78	0.33	0.80	0.22	3
C	2.98	0.47	0.00	1.02	2
D	3.24	0.21	0.26	0.76	1
E	4.0	0.55	1.02	0.00	3
F	3.9	0.45	0.92	0.10	3

Step 3 : Recalculate the cluster centers

Now, that we have assigned each data point to a cluster, we can calculate new cluster center by taking the mean of all the data points assigned to each cluster.

Cluster-1 : center, $(3.45 + 3.24)/2 = 3.345$

Cluster-2 : center, 2.98

Cluster-3 : $(4.0 + 3.78 + 3.9)/3 = 3.893$

Step 4 : Repeat steps 2 and 3 until convergence

Now we assign each student's CGPA to the newly calculated cluster centers.

Student	CGPA	Distance from Cluster-1	Distance from Cluster-2	Distance from Cluster-3	Assigned Cluster
A	3.45	0.105	0.47	0.443	1
B	3.78	0.435	0.80	0.113	3
C	2.98	0.36	0.00	0.913	2
D	3.24	0.105	0.26	0.653	1
E	4.0	0.655	1.02	0.107	3
F	3.9	0.555	0.92	0.007	3

Now, we can calculate the new cluster centers. The centers are same as previous. As we can see that the cluster centers have stopped changing after the second iteration.

So the final clusters are:

Cluster-1:

center:- 3.345 (mean CGPA)

Assigned students :- A, D

Cluster-2:

center:- 2.98 (mean CGPA)

Assigned students :- C

Cluster-3:

center:- 3.893 (mean CGPA)

Assigned students : B, E, F

19. Define the following terms:

Search space, population, Generation

Ans: The definitions are given below:

Search space: The search space is the set of all possible solutions to the problem being optimized. For example, if the problem is to find the maximum value of a mathematical function over a certain range, then search space would be that range.

Population: A population is set of candidate solutions (also called individuals or chromosomes) to the problem being optimized. The population is initialized randomly and evolves over time through selection, crossover and mutation process.

Generation: The process of evaluation, selection, recombination and mutation. A generation is a single iteration of the genetic algorithm.

20. What is chromosome?

Ans: A chromosome is a data structure that represents an individual in the population. It is actually a set of genes. A chromosome contains solution in form of genes. Each gene encodes a particular feature or characteristic of the chromosome.

21. What is fitness?

Ans: Fitness is the value assigned to an individual based on how far or close it is from the solution. It is actually a measure of how well an individual solves a problem. It is used to evaluate the quality of the candidate solutions and to guide the selection process in the genetic algorithm. Fitness is typically defined as a function that takes an individual as input and returns a numerical value that reflects the degree of success in solving the problem. The greater the fitness value the better solution it contains.

SHOT ON MI A2

22. Write the genetic algorithm and how it works.

Ans: The basic genetic algorithm is given below:

Step 1: Represent the problem variable domain as a chromosome of a fixed length, choose the size of a chromosome population N, the crossover probability P_c and the mutation probability P_m.

Step 2: Define a fitness function to measure the performance, or fitness, of an individual chromosome in the problem domain. The fitness function establishes the basis for selecting chromosomes, that will be mated during reproduction.

Step 3: Randomly generate an initial population of chromosomes of size N:

$$x_1, x_2, x_3, \ldots\ldots, x_N$$

Step 4: Calculate the fitness of each individual chromosome:

$$f(x_1), f(x_2), f(x_3), \ldots\ldots, f(x_N)$$

22. Write the genetic algorithm and how it works.

Ans: The basic genetic algorithm is given below:

Step 1: Represent the problem variable domain as a chromosome of a fixed length, choose the size of a chromosome population N, the crossover probability P_c and the mutation probability P_m.

Step 2: Define a fitness function to measure the performance, or fitness, of an individual chromosome in the problem domain. The fitness function establishes the basis for selecting chromosomes, that will be mated during reproduction.

Step 3: Randomly generate an initial population of chromosomes of size N:

$$x_1, x_2, x_3, \ldots\ldots, x_N$$

Step 4: Calculate the fitness of each individual chromosome:

$$f(x_1), f(x_2), f(x_3), \ldots\ldots, f(x_N)$$

Step 5 : Select a pair of chromosomes for mating from the current population. Parent chromosomes are selected with a probability related to their fitness.

Step 6 : Create a pair of offspring chromosomes by applying the genetic operators. crossover and mutation.

Step 7 : Place the created offspring chromosomes in the new population.

Step 8 : Repeat step 5 until the size of the new chromosome population becomes equal to the size of the initial population, N.

Step 9 : Replace the initial (parent) chromosome population with new (offspring) population.

Step 10 : Go to step 4, and repeat the process until the termination criterion is satisfied.

SHOT ON MI A2

How it works:

1. Initialization: The initial population is generated randomly, with each individual represented as a chromosome. The size of the population and the representation of the chromosome depend on the problem being solved.

2. Fitness evaluation: Each candidate solution is evaluated by applying a fitness function to the chromosome. The fitness function evaluates how well each candidate solution solves the problem being addressed by the genetic algorithm. The fitness function can take into account different aspects of the problem, such as cost, accuracy, or efficiency of the solution.

3. Selection: A subset of the candidate solutions is selected for the reproduction in the next generation. The probability of selection is proportional to the fitness of each candidate

solution. Individuals with higher fitness have a very higher probability of being selected for reproduction. There are several different selection methods, including roulette wheel selection, tournament selection, rank-based selection.

4. Reproduction: The selected candidate solutions are used to create new offspring through the process of crossover and mutation. Crossover involves combining parts of the chromosome of two parents solution to create new offspring. Mutation involves making random changes to a chromosome, such as flipping a bit in a binary string or changing a numerical value in a vector.

5. Replacement: The new offspring are added to the population, replacing some of the existing solutions. The number of offspring and the replacement strategy depend on the specific implementation of the genetic algorithm.

23. Graphically show the following most widely used tools and metrics in classification problem.

i) Confusion matrix ii) Area under ROC curve

Ans:

i) Confusion matrix: A confusion matrix is a table that is used to evaluate the performance of a classification model. It summarizes the number of correct and incorrect predictions that are made by the model, compared to the actual class labels in the data. A confusion matrix is given below:

Actual Values \ Predictions	0	1
0	TN	FP
1	FN	TP

Here,

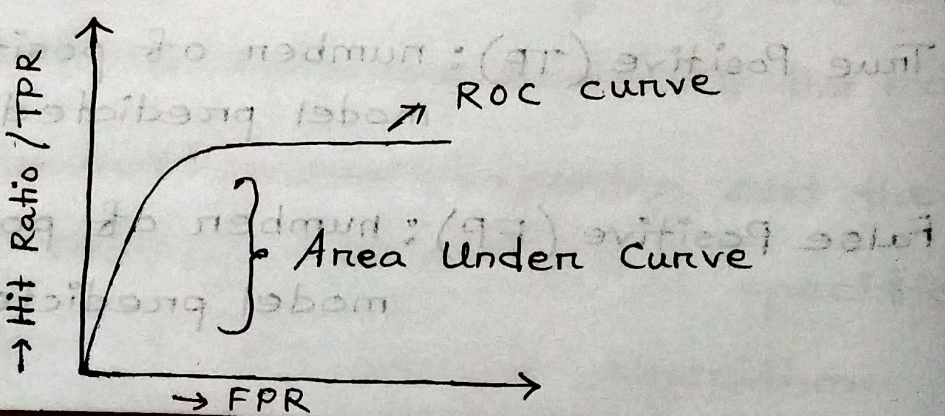
True Positive (TP): number of positive instances the model predicted correctly

False Positive (FP): number of positive instances the model predicted incorrectly

True Negative (TN): number of negative instances the model predicted correctly

False Negative (FN): number of negative instances the model predicted incorrectly

ii) Area under ROC curve: The ROC curve is a graphical representation of the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) of a binary classifier. The AUC-ROC is the area under ROC curve, and it is valued between 0 and 1. This is a popular metric for evaluating binary classification models because it is relatively insensitive to imbalanced class distribution and can be used to compare the performance of different dataset. It is useful for selecting the best threshold value.



24. What is weight and bias in perceptron?

Ans: In a perceptron, weight and bias are the two main parameters that are used to adjust the output of the model.

Weight: Weight is numerical value that is assigned to each input feature in the perceptron. It represents the strength of the connection between input and output. A positive value indicates that the feature has a positive effect on the output, while a negative weight value indicates that the feature has a negative effect on the output.

Bias: Bias is a constant term that is added to the weighted sum of the input features in the perceptron. It allows the perceptron to shift the decision boundary in the input space, and can help the model to better fit the training data.

Together, the weight and bias values in a perceptron are used to compute the weighted sum of the input features, which is passed through an activation function.

to produce the final output of the model.

25. What is CART?

Ans: Classification And Regression Trees (CART) is machine learning algorithm used for supervised learning tasks such as classification and regression. It uses a regression decision tree model to make predictions, with each branch representing a different decision or outcome. It is a non-parametric technique, meaning that it makes no assumptions about the underlying data distribution. At each node of the tree, the algorithm selects the variable and the value that best splits the data into two branches. CART models are easy to interpret and can be used to create understandable rules that can be used to make decisions.

26. Let function $f(x) = 2x - x^2/16$ be defined on the interval $[0, 31]$. Write a program to show the use of a genetic algorithm for determining the maximum of the function in the given interval. Note that, the relative fitness function $g(x)$ is defined as below:

$$g(x) = \frac{f(x)}{\sum_{x \in p}^{(k)} f(x)}$$

Ans: To begin the algorithm, we select an initial population of 4 chromosomes at random. The resulting initial population of chromosomes is shown below:

Chromsome Number	Initial Population	x value	Fitness value $f(x)$	Selection Probability
1	11010	26	9.75	0.1824
2	01110	14	15.75	0.2947
3	01100	12	15	0.2807
4	01001	9	12.9375	0.2421

$\Sigma f(x) = 53.4375$

Average = 13.35

Maximum value = 15.75 [x-value = 14]

We select the chromosomes that will reproduce on their fitness values, using the following probability:

$$P(\text{chromosome } i \text{ reproduces}) = \frac{f(x_i)}{\sum_{K=1}^{10} f(x_K)}$$

Since our population has 4 chromosomes and each 'mating' produces 2 offspring, we need 2 matings, to produce a new generation of 4 chromosomes. The selected chromosomes are displayed below. To create their offspring, a cross-over point is chosen at random, which is shown in a table as vertical line.

Chromosome Number	Mating pair	New population	x value	Fitness value f(x)
3	01|100	01010	10	13.75
1	11|010	11100	28	7
2	0111|0	01111	(15)	(15.9375) Max
4	0100|1	01000	8	12

Table: Reproduction and second generation $\Sigma f(x) = 48.6875$

27. Write some common applications of genetic algorithm.

Ans: Some applications are given below:

1. Scheduling: Genetic algorithms are often used to solve scheduling problems, such as assigning tasks to machines, finding the optimal order in which to perform tasks, and scheduling employees.

2. Data mining: Genetic algorithms are used in data mining to explore large data sets, find patterns and identify relationships.

3. Image processing and computer vision: Genetic algorithms can be used to identify objects in images, optimize image compression and segment images into regions.

4. Optimization problems: Genetic algorithms are used to solve optimization problems, such as finding the optimal route for a delivery truck or the most cost-effective way to build computer network

5. Machine learning: Genetic algorithms are used in machine learning to find the most efficient solution for a given problem.

6. Weather forecasting: Genetic algorithms can be used to make predictions about the weather, such as forecasting temperature and rainfall.

7. Feature selection: Genetic algorithms can be applied to select a subset of features that are most relevant to the problem at hand. This is particularly useful when the dataset contains a large number of features and some of them are irrelevant.

Some other applications are:

→ Data clustering
→ Bioinformatics
→ Simulated Annealing
→ Medicine
→ Robotics and
→ Filtering and signal processing etc.

28. Name and describe the main features of Genetic Algorithm (GA).

Ans: Genetic Algorithms (GA) use principles of natural evolution. There are five important features of GA:

Fitness function represents the main requirements of the desired solution of a problem (i.e. cheapest price, shortest route, most compact arrangement, etc.). This function calculates and returns the fitness of an individual solution.

Encoding possible solutions of a problem are considered as individuals in a population. If the solutions can be divided into a series of small steps then these steps are represented by genes and a series of genes (a chromosome) will encode the whole solution. This way different solutions of a problem are represented in GA as chromosomes of individuals.

Selection operations defines the way individuals in the current population are selected for reproduction. There are many strategies for that (e.g. roulette-wheel, ranked, tournament selection, etc.), but usually the individuals which are more fit are selected.

Crossover operator defines how chromosomes of parents are mixed in order to maintain/obtain genetic codes of their offspring (e.g. one-point, two-point, uniform crossover, etc.). This operator implements the inheritance property (offspring inherit genes of their parents).

Mutation operator creates random changes in genetic codes of the offspring. This operator is needed to bring some random diversity into the genetic code. In some cases GA cannot find the optimal solution without mutation operator (local maximum problem).

Year - 2020 (Extra Note)

29. Write short notes on: i) Artificial Neurons
ii) Artificial Neural Network

Ans: i) Artificial Neurons: Artificial Neurons are computational models inspired by the structure and function of a biological neurons found in the human brain. They form the basis of Artificial Neural Networks (ANNs), a powerful machine learning technique used to create predictive models that are able to learn from data and identify patterns. Artificial Neurons are composed of three main components:

→ An input layer
→ An output layer
→ One or more hidden layers

Each neuron in an ANN is connected to other neurons through weighted links, which can be adjusted to modify the strength of the connection between them. The neurons then interact by passing signal between them.

ii) Artificial Neural Networks: ANNs are computing systems inspired by biological neural networks that constitute human brains. Such systems learn to perform tasks by considering examples, generally without being programmed with any task-specific rules.

ANNs are composed of interconnected processing units (neurons) organized into layers. Each neuron in the network receives input signals from some of the neurons in the previous layer, performs mathematical operations on those signals, and sends an output signal to some of the neurons in the next layer. The connections between neurons are referred to as weights, and the strength of each connection is determined by the learning process. The goal of the learning process is to adjust the weights so that the network produces the correct output for a given set

of inputs.

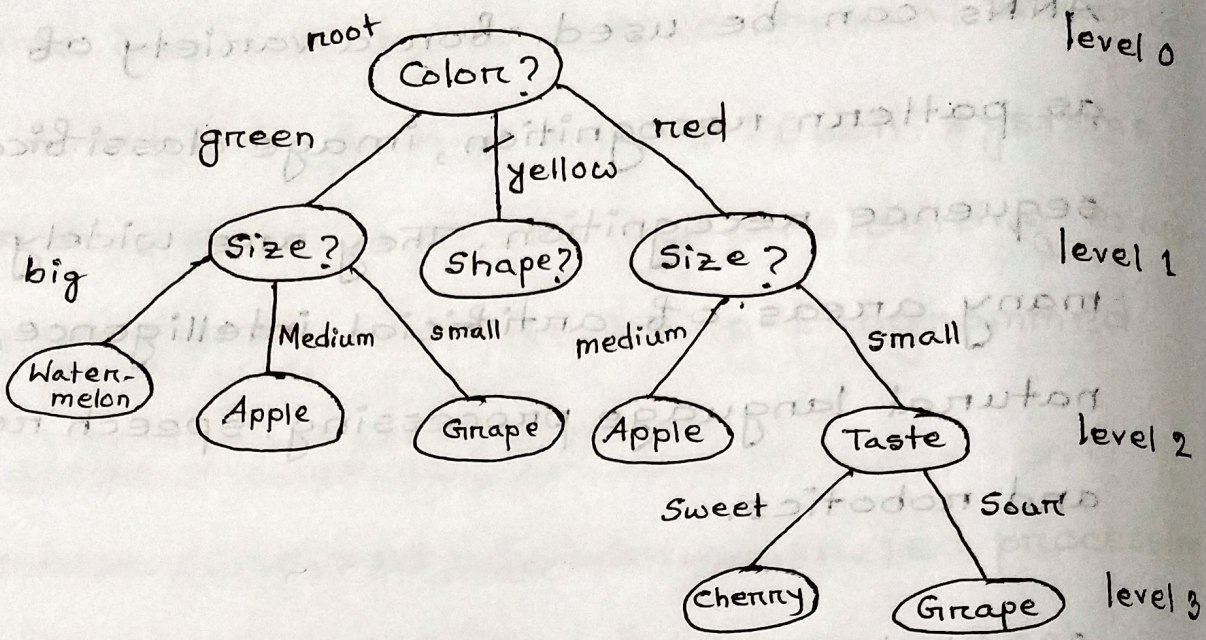
ANNs can be used for a variety of tasks, such as pattern recognition, image classification, and sequence recognition. They are widely used in many areas of artificial intelligence, such as natural language processing, speech recognition, and robotics.

30. Write short notes on Decision Tree.

Ans: A decision tree is a popular machine learning algorithm that is used for both classification and regression tasks. The basic idea behind a decision tree is to recursively, partition the input data into subsets based on the values of input features, until a stopping criterion is met. The resulting tree consists of nodes that represents decision points, and edges that represent the possible outcomes of each decision. The leaves of the tree correspond to the final decision or prediction.

SHOT ON MI A3

A simple decision tree is given below:



The process of constructing a decision tree involves two main steps : tree induction and tree pruning. In tree induction, the algorithm selects the best feature to split the data based on a criterion such as the Gini impurity or the information gain. In tree pruning, the algorithm simplifies the tree by removing nodes that don't contribute significantly to the classification or prediction accuracy. They are easy to interpret and explain. But they prone to overfitting if the tree is to deep or if the training data is noisy.

31. Describe four important string problems in pattern recognition.

Ans: There is a large number of problems in computations on strings. The ones that are of greatest importance in pattern recognition are:

1. String matching: This problem involves finding the occurrence(s) of a given pattern (or string) within a larger text (or string). Given x and text, test whether x is a factor of text, and if so, where it appears. There are several algorithms for solving this problem, such as brute-force, Knuth-Morris-Pratt (KMP), Boyer-Moore(BM), and Rabin-Karp etc.

2. Edit distance: This problem involves finding the minimum number of operations (insertion, deletion, substitution) required to transform one string to another. Given two strings x and y, compute the minimum number of basic operations needed to transform x into y. The most com-

monly used measure of edit distance is Levenshtein distance.

3. String matching with errors: In this problem, the goal is to find all occurrences of a pattern string in a text string, allowing for a certain number of errors or mismatches between the pattern and text. The errors can include substitutions, deletions, and insertions of characters in the pattern string. Given x and text, find the locations in text where the "cost" or "distance" of x to any factor of text is minimal.

4. String matching with the 'don't care' symbol: In this problem, the goal is to find all occurences of a pattern string in a text string, allowing for a certain positions in the pattern to match any character in the text, represented by a 'don't care' symbol (usually denoted by a ∅ or ?). This is the same as basic

string matching, but with a special symbol, ϕ, the don't care symbol, which can match any other symbol. The solution to this problem involves using algorithms such as the Aho-Corasick algorithm or the finite automaton to construct a DFA that recognizes all possible matches of the pattern in the text, including matches that contain the 'don't care' symbols.

32. Write down Boyer-Moore string matching algorithm and explain with an example.

Ans: The Boyer-Moore string matching algorithm is given below:

```
begin initialize A, x, text, n ← length [text], m ← length [x]
      F(x) ← last-occurrence function
      G(x) ← good-suffix function
      s ← 0
      while s ≤ n-m
            do j ← m
            while j > 0 and x[j] = text [s+j]
```

do $j \leftarrow j-1$

if $j=0$

then print "pattern occurs at shift" s

$s \leftarrow s + G(0)$

else $s \leftarrow s + \max\left[G(j), j - F(\text{text}[s+j])\right]$

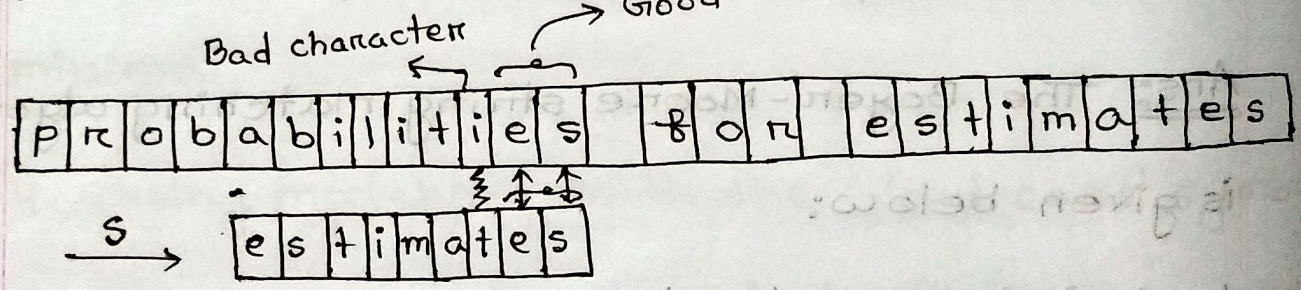
return

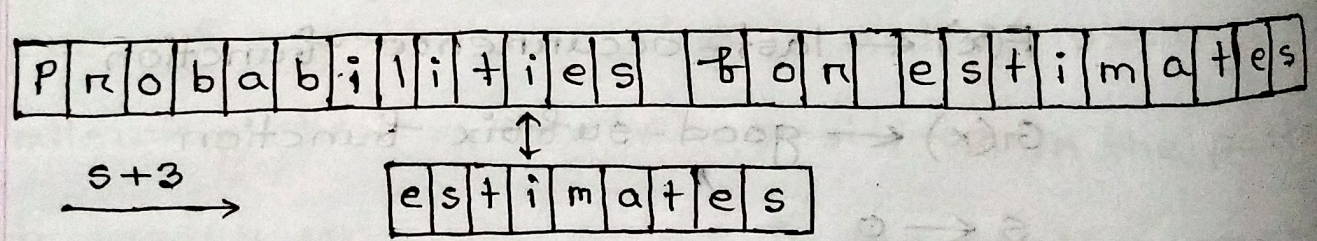
end

An Example is given below:

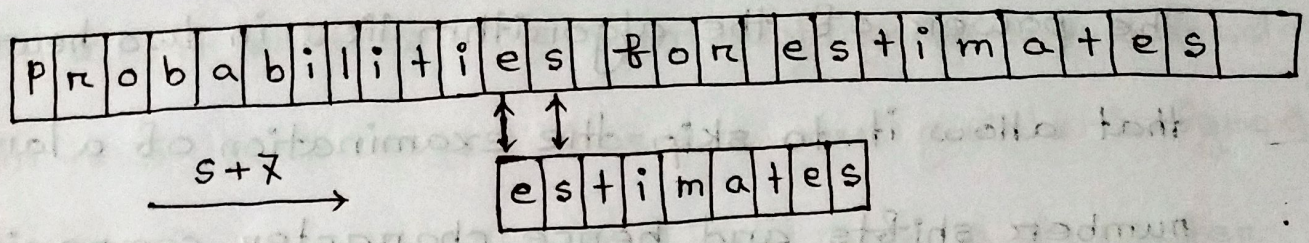
text : probabilities for estimates

pattern: estimates





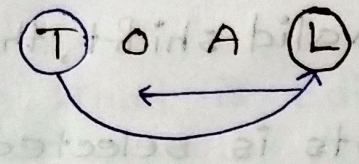
proposed by bad character heuristic



proposed by good
suffix heuristic

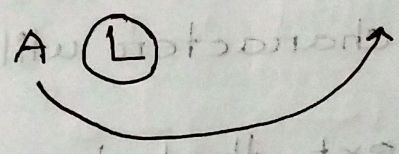
Another Example:

0	1	2	3	4	5	6	7	8	9	10	11	12
W	E	L	T	O	M	E	T	O	A	L	G	O



T exist in the pattern so move till mismatch become match

0	1	2	3	4	5	6	7	8	9	10	11	12
W	E	L	T	O	M	E	T	O	A	L	G	O



Doesn't exist in the pattern so move past mismatch

0	1	2	3	4	5	6	7	8	9	10	11	12
W	E	L	T	O	M	E	T	O	A	L	G	O

T O A L

Boyer-Moore Algorithm explanation:

The power of the algorithm lies in two heuristics that allow it to skip the examination of a large number shifts and hence character comparisons: the good-suffix heuristic and the bad-character heuristics operate independently and in parallel. After a mismatch is detected, each heuristic proposes an amount by which s can be safely increased without missing a valid shift, the larger of these proposed shifts is selected and s is increased accordingly.

The bad character utilizes the rightmost character in text that doesn't match the aligned character in x. It proposes incrementing the shift by an amount to align the rightmost occurrence of the bad character in x with the bad character identified in text.

A general suffix of x is a factor or substring of x that contains the final character

in x. At shift s the rightmost contiguous characters in text that match those x are called the "good suffix" or "matching suffix". Once a character mismatch has been found, the good-suffix heuristic proposes to increment the shift so as to align the next character/occurrence of the good suffix in x with that identified in text.

33. What is Edit distance? Write down Edit distance algorithm and explain with an example.

Ans: Edit Distance: Edit distance, also known as Levenshtein distance, is a measure of the difference between two strings, defined as the minimum number of operations required to transform one string into another. The allowed operations are usually defined as insertions, deletions, and substitution of individual characters.

For example, the edit distance between the strings 'cat' and 'cot' is 1, since only a single substi-

tution (changing 'a' to 'o') is needed to transform one string into the other.

Edit Distance Algorithm:

begin intialize A, x, y, m ← length [x], n ← length [y]

c[0,0] ← 0

i ← 0

do i ← i+1

c[i,0] ← i

until i=m

do j ← j+1

c[0,j] ← j

until j=n

i ← 0 j ← 0

do i ← i+1

do j ← j+1

Insertion ↗ Deletion ↗

$$c[i,j] = \min\left[c[i-1,j]+1,\ c[i,j-1]+1,\ c[i-1,j-1] - \delta(x[i], y[j])\right]$$

↙ no change / exchange

until j=n

until i=m

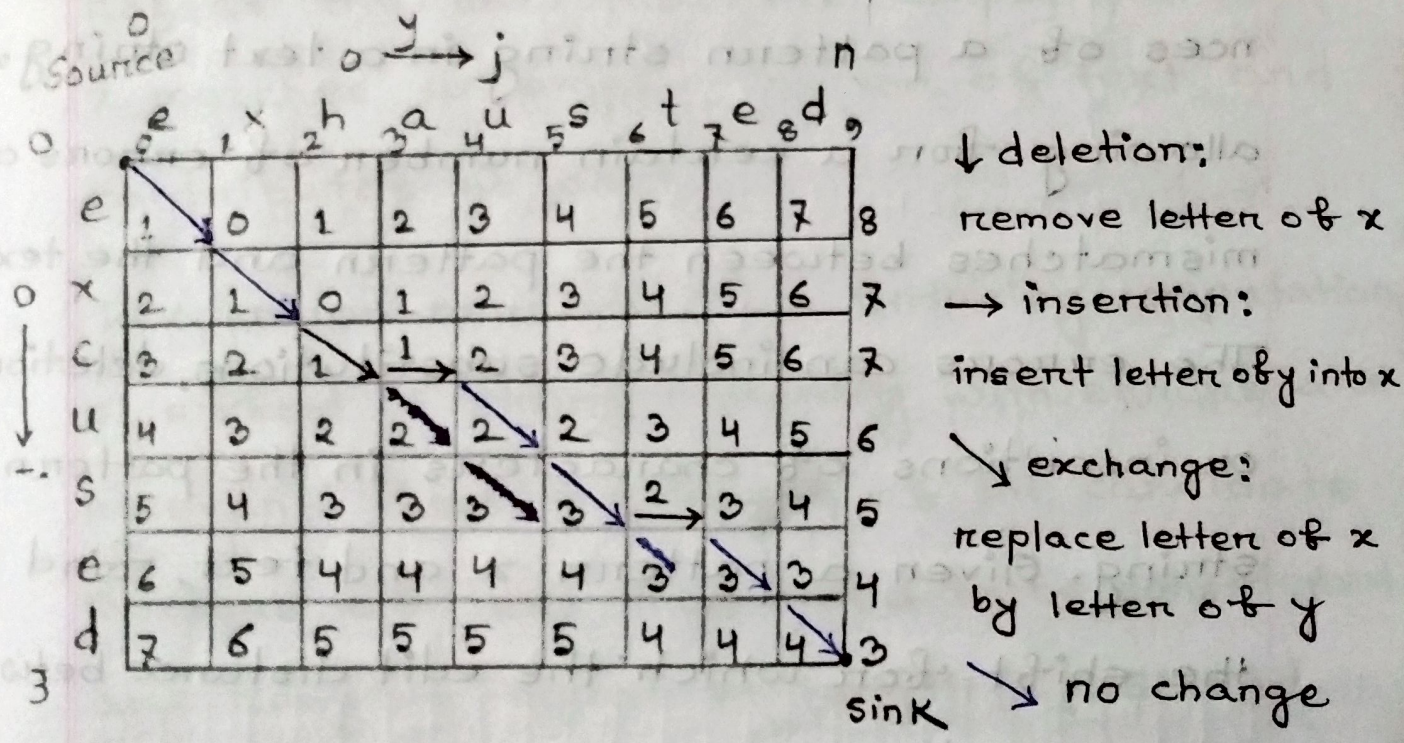
return c[m, n]

end

An Example:

Let, first string, x = "excused"

second string, y = "exhausted"



x	excused	source string	C[0,0] = 0
	exhused	substitute c→h	C[3,3] = 1
	exhaused	insert a	C[3,4] = 2
	exhausted	insert t	C[5,7] = 3
y	exhausted	target string	C[7,9] = 3

34. Write short notes on:

i) string matching with errors

ii) string matching with the 'don't care' symbol

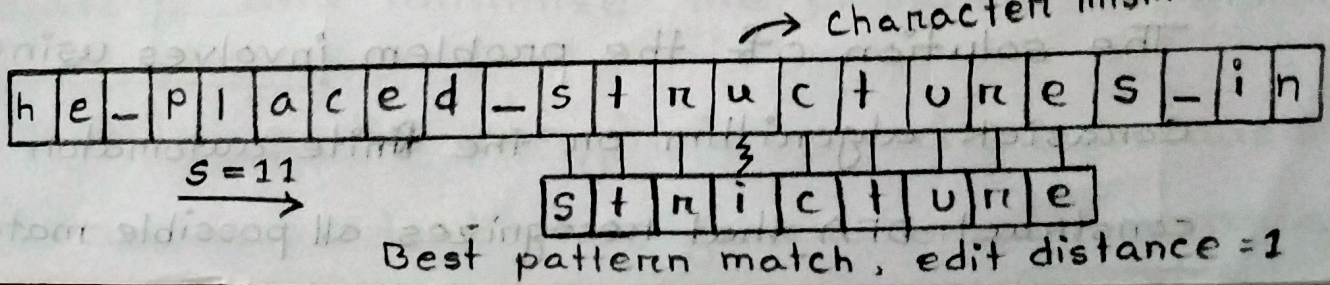
Ans: i) String matching with errors:

In this problem, the goal is to find all occurrences of a pattern string in a text string, allowing for a certain number of errors or mismatches between the pattern and the text. The errors can include substitutions, deletions, or insertions of characters in the pattern string. Given a pattern x and text, find the shift for which the edit distance between x and a factor of text is minimum. The algorithm for string matching with errors is very similar to that for edit distance. We seek a shift for which the edit distance to a factor of text is minimum, or formally $\min[C[x,y])$, where y is

any factor of text.

The principal difference of the algorithms is that we initialize E[0,j] to 0 in string matching with errors instead of j of the basic string matching problem. This initialization of E expresses the fact that the 'empty' prefix of x matches an empty factor of text, and contributes no cost.

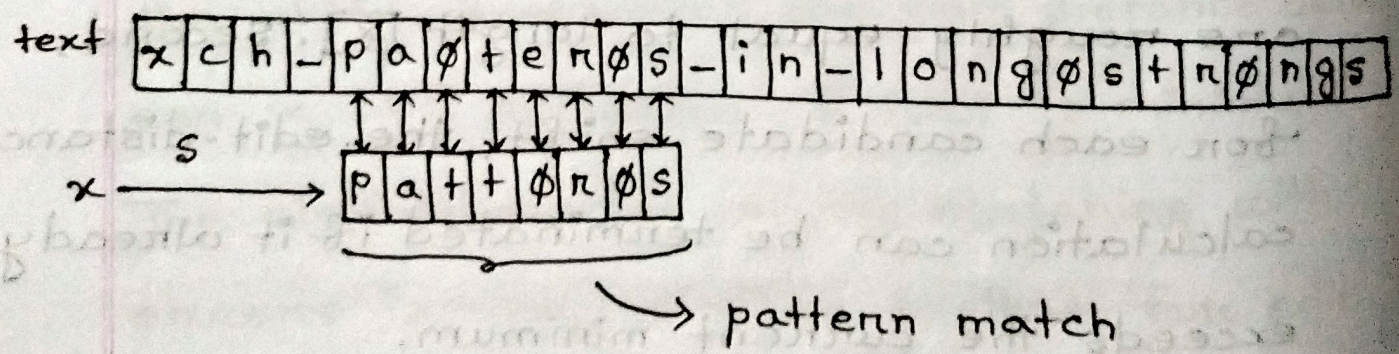
Two minor heuristic for reducing computational effort in string matching with errors are relevant. First, the length of the candidate factors of text that need to be considered are roughly equal to length[x]. Second, for each candidate shift, the edit-distance calculation can be terminated if it already exceeds the current minimum.



ii) String matching with the 'Don't-care' Symbol:

In this problem, the goal is to find all occurrences of a pattern string in a text string, allowing for certain positions in the pattern to match any character in the text, represented by a 'don't care symbol' (usually denoted by a '?' or 'ϕ'. This is the same as basic string matching, but with a special symbol, the don't care symbol, which can match any other symbol.

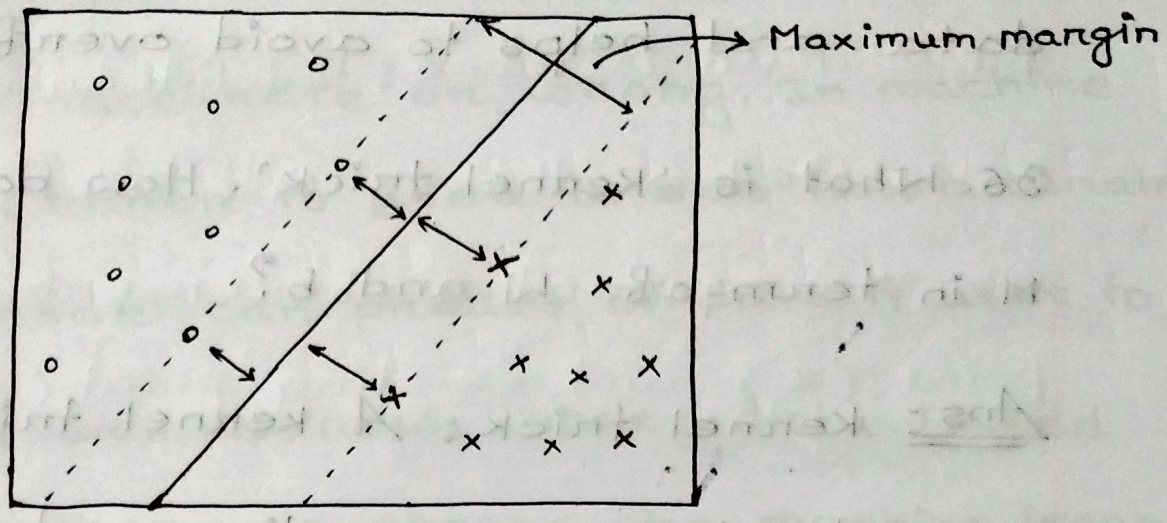
An Example is given below:



The solution of the problem involves using Aho-Corasick algorithm or the finite automaton to construct DFA that recognizes all possible matches.

35. Why maximum margin in SVM technique?

Ans: Maximum margin in SVM technique:



The maximum margin in SVM technique is important because it provides a way to find the best decision boundary between two classes.

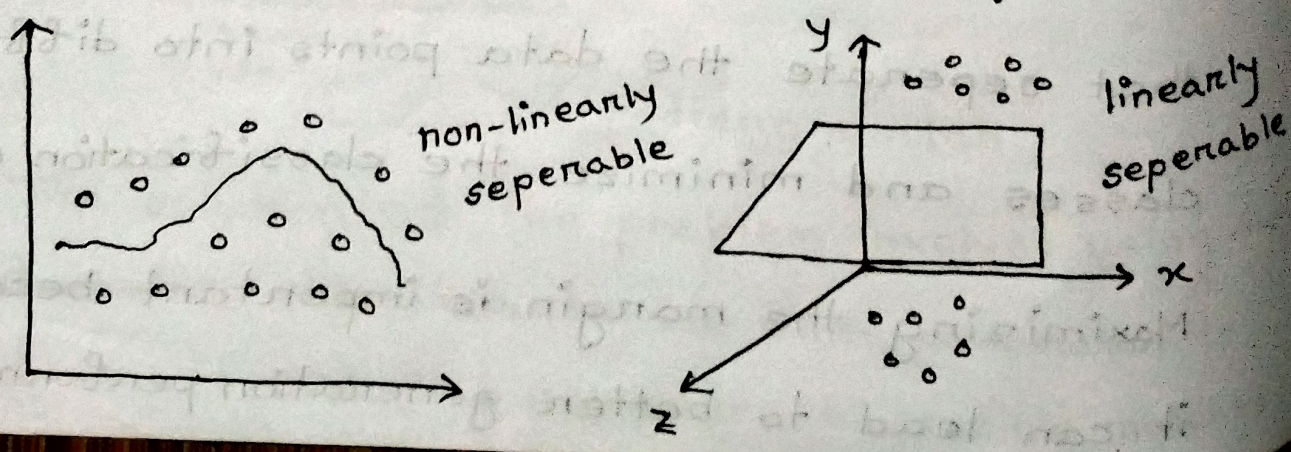
In SVM, decision boundary is a hyperplane that seperate the data points into different classes and minimize the classification error.

Maximizing the margin is important because it can lead to better generation performance

The larger the margin, the more confident we can be that the decision boundary is correct. It will perform better for new data and helps to avoid overfitting.

36. What is 'Kernel trick'. How do you compute M in term of W and b?

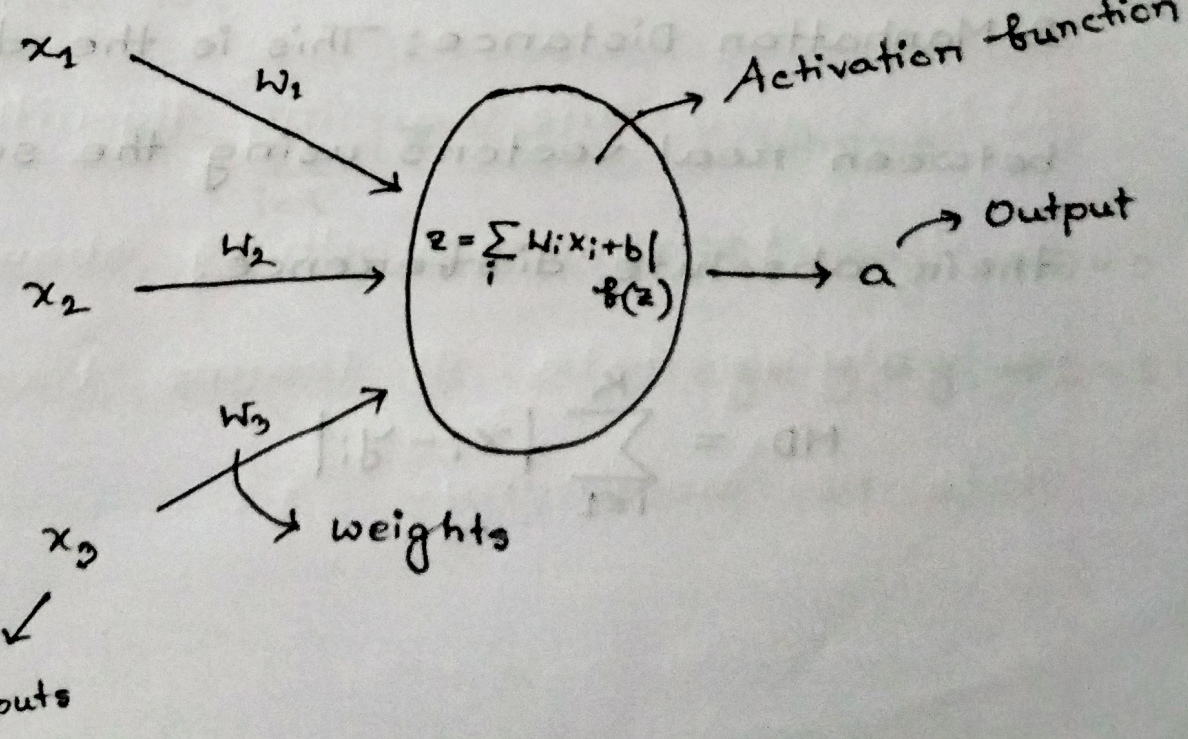
Ans: Kernel trick: A kernel trick is a simple method where a non-linear data is projected onto a higher dimension space so as to make it easier to classify the data where it could be linearly divided by a plane. This is mathematically achieved by Lagrangian formula using Lagragians multipliers.



37. Define error.

Ans: The definition of error is given below:

Error: Error can be described as an action which is inaccurate or wrong. In machine learning, error is used to see how accurately our model can predict on data it uses to learn as well as new, unseen data. Based on our error, we choose the machine learning model which performs best for a particular dataset.



38. What are distance method used in KNN algorithm?

Ans: The distance method are discussed below:

1. Euclidean Distance: Euclidean distance is calculated as the square root of the sum of the squered difference between new point (x) and existing point (y).

$$ED = \sqrt{\sum_{i=1}^{K} (x_i - y_i)^2}$$

2. Manhattan Distance: This is the distance between real vectors using the sum of their absolute difference.

$$MD = \sum_{i=1}^{K} |x_i - y_i|$$

3. Minkowski Distance: It is the general form of Euclidean and Manhattan distance. It can be calculated as follows:

$$D = \sum_{i=0}^{n} |P_i - Q_i|^{1/p}$$

4. Hamming Distance: It is used for categorical variables. If the value (x) and the value (y) are the same, the distance D will be equal to 0, otherwise D = 1. The general formula is:

$$D = \sum_{i=1}^{K} |x_i - y_i| \quad ; \text{ where,}$$

if $x = y, \Rightarrow D = 0$

else $x \neq y, \Rightarrow D = 1$

39. What is KNN algorithm?

Ans: K-nearest neighbors is a type of supervised machine learning algorithm which can be used for both classification as well as regression predictive problems.

There are two main problems.

i) Lazy learning algorithm: Because it doesn't have a specialized training phase and uses all the data for training while classification.

ii) No parametric learning Algorithm: As it doesn't assume anything about the underlying data. It learns from training data not assuming.

40. What are the advantage and disadvantage of cross-validation?

Ans: Cross validation is a technique that is used to detect overfitting. It has some advantages and disadvantages which are given below:

Advantage:

→ less dependent on how data gets divided

→ Every data point gets to be in a test set exactly once and gets to be in training set (K-1) times, if the dataset is divided into K parts.

Disadvantage:

→ It requires a lots of time: that means cross validation is time-consuming.

SHOT ON MI A2
MI DUAL CAMERA